SOFTWARE CHANGE IMPLEMENTATION REPORT

CR-010: Student Profile Image Upload and Management

Field	Details
Change Request ID	CR-010
Change Request Title	Student Profile Image Upload for the Admin Student Management
Student Name	P.Tharanan
Student ID	JF/ICT/24/20
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Currently, administrators can add and manage student information such as registration number, name, email address, phone number, and other existing student details. However, the system does not provide an option to upload or display a student's profile image. When managing a large number of students, a profile image can help administrators identify students more easily and improve the usability of the student management system. Therefore, the system should allow administrators to upload a profile image when adding a new student. The system should support commonly used image formats such as PNG, JPG, and JPEG. The uploaded image should be associated with the correct student and displayed in the student list or student details section. Existing Behaviour

Before implementation:

- The administrator can access the student management section.
- The administrator can add a new student using the existing Add Student form.
- Existing student information such as name, registration number, email, and phone number can be entered.
- The Add Student form does not currently contain a profile image upload field.
- Students are displayed without profile images.
- No image file validation is currently available.
- Student records are stored without a profile image path.
- Existing student CRUD operations use the existing student_id.

1.2 Requested Change

Add a Student Profile Image Upload facility to the administrator's Add Student form.

The implementation must:

- Provide a profile image upload field in the Add Student form.
- Allow administrators to select a student image from their computer.
- Support PNG, JPG, and JPEG image formats.
- Reject unsupported file formats.
- Validate the selected file before uploading.
- Display a suitable validation message when an unsupported file is selected.
- Store the uploaded image securely on the server.
- Store the image path/reference with the corresponding student record.
- Display the uploaded student image in the student list or student details page.
- Preserve the existing Add Student functionality.
- Preserve existing Edit and Delete functionality.
- Ensure the uploaded image is associated with the correct student_id.
- Maintain existing administrator authentication and authorization.
- Prevent unsafe file types from being uploaded.
- Provide a suitable default/placeholder image when no profile image is available.

2. Implementation

2.1 Implementation Summary

CR-010 implements optional student profile image management across the administrator dashboard and student portal.

Administrators can select a profile image while creating or editing a student. PNG, JPG, and JPEG files are accepted up to 10 MB; the browser and server validate the file type and image signature.

Images are stored under backend/uploads/students with a username-based filename and three-digit suffix. The image URL is saved in users.profile_img_url and returned with student records for display in the admin list and student profile.

2.2 Affected Files

File	Change	Purpose
backend/src/middleware/profileImageUpload.js	Added	Validates PNG/JPG/JPEG extension, MIME type, image signature, and 10 MB limit.
backend/src/controllers/studentController.js	Modified	Creates, replaces, and removes images with transaction-aware cleanup; adds student self-service handlers.
backend/src/models/studentModel.js	Modified	Reads image URLs from users and provides user image URL updates.
backend/src/routes/studentRoutes.js	Modified	Protects admin routes and adds STUDENT-only self-service image endpoints.
backend/src/app.js	Modified	Serves files from the uploads directory.
backend/package.json and package-lock.json	Modified	Adds multer multipart upload middleware dependency.
frontend/src/components/StudentFormModal.jsx and frontend/src/services/api.js	Modified	Sends optional images as multipart form data and validates selection.
frontend/src/pages/DashboardPage.jsx and frontend/src/styles/dashboard.css	Modified	Displays student images beside names in the admin list with an initials fallback.
frontend/src/pages/StudentDashboardPage.jsx and frontend/src/styles/studentDashboard.css	Modified	Adds avatar edit/remove actions and a validated student image dialog.

Other modified files and the profile-image management flow are listed in the table above.

2.3 Technical Changes

Profile Image Validation

The upload accepts PNG, JPG, and JPEG files only.

- The file extension and MIME type must match an accepted image format.
- The backend checks the PNG or JPEG file signature rather than trusting the browser filename alone.
- Files larger than 10 MB are rejected by the server; the form also provides an immediate size check.

Profile image upload is optional for student creation and editing.

File Storage and Database Association

The uploaded image is written to the backend student upload directory:

- backend/uploads/students
- The stored filename uses the student username (registration number), a three-digit suffix, and the original accepted extension.
- The public file reference is stored in users.profile_img_url.
- Student list and profile queries join users to return the associated profile image URL.
- A student without an image continues to display an initials placeholder.

Replacement and Removal

On administrator edit or student self-service edit, the new image is written first and the database URL is committed transactionally.

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After a successful replacement, the previous managed image is removed from
disk.
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Student self-service removal clears users.profile_img_url and removes the managed image file.

Authorization and Existing Functionality

Administrator student create/edit uploads remain behind ADMIN authentication and authorization.

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Student image edit/remove endpoints require STUDENT authorization and
derive the target user from the authenticated token.
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Existing student fields and CRUD routes remain available; student self-service cannot select another student account.

2.4 Implementation Logic

1. Load student records with their profile_img_url from the users table.
2. Render the image beside the student name in the administrator list, or initials when no URL exists.
3. Show the image controls on the student dashboard avatar; the edit control opens the validated file-selection dialog.
4. Send multipart form data to the authenticated profile-image endpoint.
5. Save the generated image file and URL for the authenticated student.
6. After a successful update, remove the previous image and refresh the displayed avatar.
7. For image removal, clear the database URL and delete the previous managed file.
8. Keep create/edit image fields optional; validate a file only when one is selected.
9. Preserve administrator/student role boundaries and existing student CRUD behavior.

2.5 Error and Data Handling

Upload middleware rejects unsupported extension/MIME combinations, mismatched image signatures, and files over the configured size limit.

Create/edit requests without an image continue to work; replacement failures leave the previous image reference and file in place.

File cleanup is restricted to paths inside backend/uploads/students and occurs only after the database update succeeds.

3. Testing & Verification

3.1 Testing Approach

Verification included a production frontend build, backend syntax checks, direct database schema inspection, and execution of the admin-list and student-profile SELECT queries. The supplied UI screenshots were reviewed; upload persistence and file cleanup were not exercised end-to-end in this session.

Checks performed:

- Frontend production build
- Backend JavaScript syntax checks
- Database profile_img_url column and joined query checks
- Screenshot review of admin list/add/edit and student image dialog
- Source review of validation, authorization, and cleanup paths
- End-to-end upload/replacement/removal with real multipart requests
- Existing CRUD regression test after image operations

Environment: Development
Browser: Chrome

Test Role: Administrator, student

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC01	Frontend production build	Vite completes successfully	Build completed successfully	Pass
TC02	Backend syntax checks	Changed backend modules parse successfully	Node syntax checks completed successfully	Pass
TC03	Database schema	users.profile_img_url exists	Column confirmed in configured MySQL database	Pass
TC04	Admin list and student profile queries	Joined queries return student rows without ambiguity	Admin list returned 2 rows; student profile query returned 1 row	Pass

TC05	Admin add/edit and student image dialog UI	Image controls and file selection are available	Supplied screenshots show list, add/edit forms, and student image dialog	UI reviewed
TC06	Invalid format and over-limit rejection	Unsupported/mismatched files and files over 10 MB are rejected	Frontend and server checks verified in code; endpoint not exercised	Code checked
TC07	Image persistence, replacement and removal	URL is stored; old file removed only after success	Implemented; end-to-end multipart/file cleanup not run	Not run
TC08	Student ownership and role authorization	Student can change only own image; admin routes remain protected	Route/controller enforce authenticated role and user ID	Code checked
TC09	Existing CRUD after image changes	Create/edit/delete continue to work	No separate CRUD regression run	Not run

3.3 Test Summary

Test Result	Count
Total checks	9
Passed	4
Code checked / UI reviewed	3
Not run end-to-end	2

Overall Test Result: Partially Verified (end-to-end upload and CRUD regression checks remain outstanding).

4. Acceptance Criteria Verification

ID	Acceptance Criterion	Verification	Status
AC-001	Administrator can select an optional image during add/edit.	Add/edit form and multipart request implementation	Implemented
AC-002	PNG, JPG, and JPEG images up to 10 MB are accepted.	Frontend and backend validation checks	Implemented; E2E pending
AC-003	Unsupported or mismatched file content is rejected.	Extension, MIME, and image signature checks	Implemented; E2E pending
AC-004	The image URL is stored on the associated user and shown in the list/profile.	Joined student queries and UI rendering	Implemented; query/build verified
AC-005	Replacing an image removes the previous managed file after commit.	Controller transaction and cleanup path	Implemented; E2E pending
AC-006	A student can edit/remove only their own image.	STUDENT route authorization and token user ID	Implemented; E2E pending
AC-007	Removing an image clears its URL and managed file.	Controller/model removal path	Implemented; E2E pending
AC-008	Existing student data and role restrictions are preserved.	Existing routes and list/profile queries	Partially verified

5. Evidence

Four evidence items are specified below. The screenshot files were referenced in the chat but are no longer available in the local attachment cache, so labeled insertion slots are retained.

Evidence 01 - Administrator Student List

Related: AC-001, AC-002, AC-003, AC-004

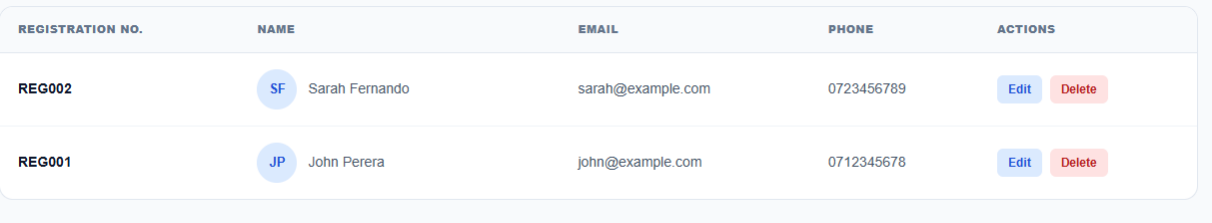


Figure 1: Administrator student list displaying student avatars beside names.

Evidence 02 - Administrator Add Student Form

Related: AC-005, AC-006, AC-007

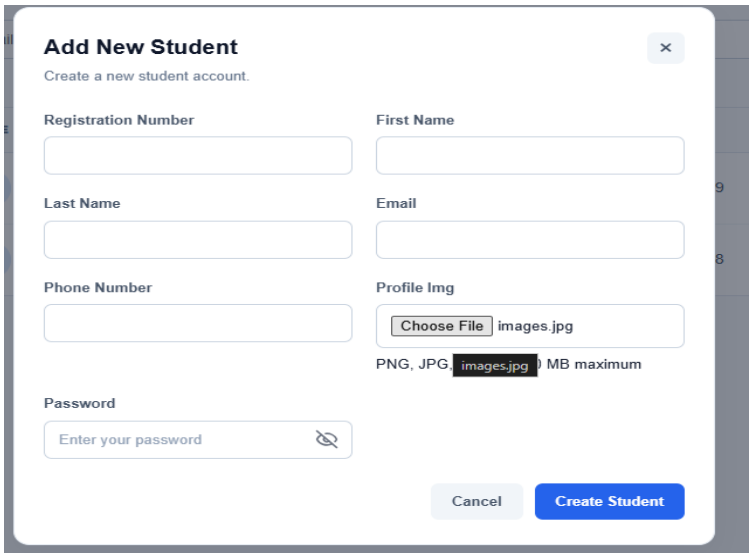


Figure 2: Add Student form with the optional Profile Img field and a selected JPG file.

5.1 Evidence 03 - Administrator Edit Student Form

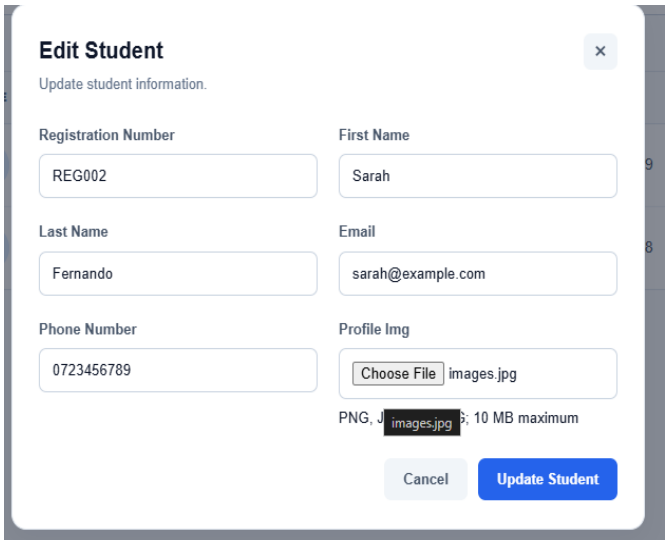


Figure 3: Administrator selects a replacement profile image while editing a student.

5.2 Evidence 04 - Student Profile Image Controls

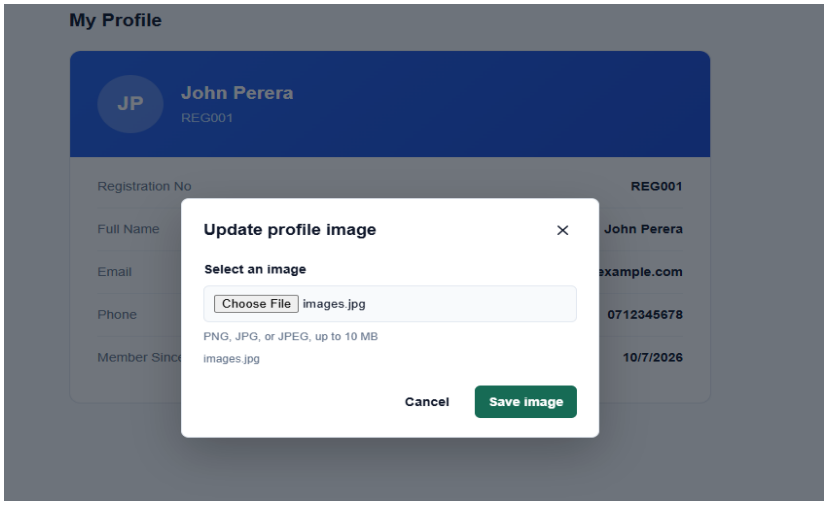


Figure 4: Student opens the profile-image dialog from the dashboard avatar controls.

6. Final Result

Item	Result
Requirements	Implemented; end-to-end verification pending
Implementation	Completed
Testing	Partially verified
Acceptance Criteria	Implemented; final tests pending
Existing CRUD	Not fully regression-tested
Final Status	Pending final upload/removal verification

6.1 Final Summary

CR-010 adds optional profile image upload to administrator student creation and editing, shows stored images in the admin student list, and gives students authenticated controls to update or remove their own image. PNG/JPG/JPEG type and signature checks, the 10 MB limit, username-based filenames, users.profile_img_url persistence, and post-commit file cleanup are implemented. Frontend build, backend syntax, database schema, and joined-query checks passed. Full multipart upload, replacement, removal, and CRUD regression tests remain to be run.

6.2 Issues / Limitations

End-to-end HTTP tests for valid/invalid image uploads, database persistence, replacement cleanup, and student removal were not run during this report preparation.

If none:

The four screenshot files are not embedded because their pasted-image cache files are unavailable. Reattach the original images to replace the labeled evidence slots.

Student Name: ___P.Tharanan____

Student ID: JF/ICT/24/20

Date: 13 October 2026

Signature: _____P.Tharanan_____

~ END OF REPORT ~